Driving Assistance of Welfare Vehicle with Virtual Platoon Control Method which has Collision Avoidance Function using Mixed Reality

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***Abstract*—Recently, welfare vehicles are widely used among the elderly and the aged. However, it is not easy for inexperienced beginners or the aged to control the welfare vehicle in narrow places such as elevators and corridors. In this paper, a driving assistance using MR which has collision avoidance function is proposed using a virtual platoon control. In the system, the cognitive and driving ability is enhanced from the simulated objective view point. The driver rides on the vehicle and controls using the virtual preceding vehicle with a Head Mounted Display (HMD). The virtual preceding vehicle projected on a HMD steered virtually from a simulating objective viewpoint, and the real vehicle is robust controlled so as to follow the virtual preceding vehicle. Furthermore, the obstacle avoidance function in the narrow space is implemented on the assistance system with MR technology. The feature of the proposed assistance system is a modification of the driver's operation not to collide the static obstacles using 3D modeling function of the environment by HMD. As a result, it is confirmed that collision can be avoided safely by correction for the driver's operation of the virtual preceding vehicle control.**

***Index Terms*—mixed reality, virtual platoon control, welfare vehicle, objective viewpoint, obstacle avoidance**

I. Introduction

In an aging society, the activity range of the elderly and the aged will be expanded and their quality of life will be greatly improved because many people who need assistance use welfare vehicles. In wide places, the aged can drive the vehicles well even though the inexperienced aged are the drivers. In narrow places, the collision with the wall and the obstacle occurs sometimes. The reason for the accidents seems to lack of the driving skill and/or the cognitive capacity of the driving environment. For the aged, the driving assistance system to be safer is required that enhances the driving skill and the cognitive capacity simultaneously.

It is known that the control with an objective (third-person) viewpoint is easier than one of subjective viewpoint (first-person) viewpoint in video games. Because it is possible to understand the environment objectively while viewing the surrounding and my own image, it is easy to remote-control the machine for beginners with an objective viewpoint [1]. For example, objective viewpoint by the around view camera is sometimes used in the back driving from the overhead view. From the safety driving, however, this method cannot be used because the gap between the real field view and overhead view is so large. In fact, the remote control with an objective viewpoint using a monitor requires complex processing such as a viewpoint conversion method and its processing delay caused by the viewpoint conversion make the system hazardous [2] [3].

For the problem, a practical assistance method has been proposed using a virtual platoon control method for a welfare vehicle STAVi [4]. The virtual platoon control consists of an unmanned virtual preceding vehicle and a real following vehicle that driver rides on. Augmented Reality (AR) was used for an effective remote control of the virtual preceding vehicle. Using this method, the vehicle was driven from a simulated objective viewpoint and the effectiveness of the method was evaluated by driving experiments [5]. From the experiment, however, the beginner drivers sometimes collided with the wall because of skill shortage. So, the Mixed Reality (MR) to model the 3D-driving environment and the more safety system which avoids the collision with the wall are required.

In this paper, a driving assistance using MR which has collision avoidance function is proposed using a virtual platoon control from a simulated objective viewpoint. First, the concept of the virtual platoon by MR is shown, and the design method of precise platoon control is outlined. In the preliminary experiment, although the traveling route initially collides with obstacles, it gets closer to the expert's trajectory every time. Since the reason for driving failure was a collision with the wall, an avoidance algorithm of static obstacles is implemented into the assistance system using 3D modeling the static environment using MR. The proposed system is assessed by trial runs using STAVi.

II. Concept of virtual platoon control

The concept of our method is to drive the vehicle using virtual platoon and to use the control using 3D modeling of a running environment obtained by HMD. This is very simple but effective assist method with a combination of an advanced control theory and an image processing.

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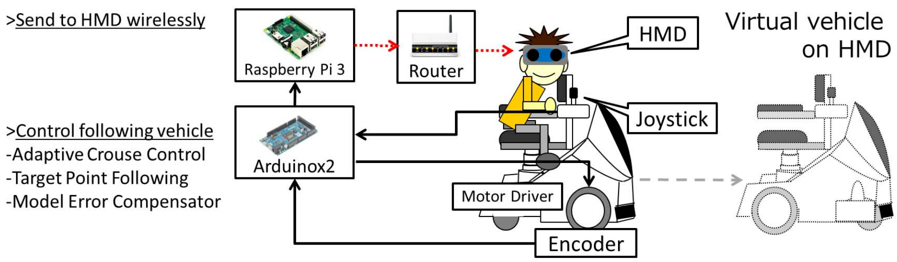


Fig. 1. Virtual platoon scheme using MR and its experimental system

The assist method is as follows; a virtual vehicle is projected using MR, and the vehicle is steered from the following vehicle looking-from-behind view-point. The proposed scheme is shown in Fig.1. The major characteristic is to simulate an objective viewpoint by immersion. In the figure, the viewpoint does not move backside, but an objective viewpoint can be simulated by setting the preceding vehicle expressly.

The virtual platoon consists of a virtual preceding vehicle and a following vehicle. The driver wears a Head Mounted Display (HMD: HoloLens) and the vehicle is controlled by the control boards. The virtual vehicle is projected on HMD, and the driver controls the virtual vehicle looking-from-behind view-point with the joystick remotely.

Fig. 2. Experimental vehicle STAVi and HoloLens

MR can be realized by superimposing a virtual object such as CG on the image of the real world. In order to display a virtual vehicle and to superimpose on a real image, the projection point must be calculated in real time. For example, it is necessary to acquire not only the position/attitude of the driver but also ones of the following vehicle where the driver wears HMD. Fig.2 shows the experimental vehicle STAVi [6] and HoloLens. Here, HoloLens is developed by Microsoft is used as a Mixed Reality device. As the driver can see the target through HoloLens, the driver will be able to stop the vehicle immediately even in dangerous situations.

The control system consists of two controllers; one is motion controllers and the other is a mixed reality controller. In motion controller, Atmel AVR controller calculates the speed, the steering angle, the position/attitude of the following vehicle are controller. And the information is sent to HMD which gives the AR image to a driver via the router using the wireless LAN. The visual image from a simulated objective viewpoint is created by combining with the virtual vehicle and landscape which is visible though HMD, where the head pose of the driver is estimated by an Inertial Measurement Unit (IMU) on HMD. In mixed reality controller, HoloLens projects a virtual image and builds a 3D-model of the environment. Where, 3D-modeling method using MR is discussed in section 5.

III. Robust platoon control with a virtual preceding vehicle

The preceding virtual vehicles are operated remotely by the joystick looking-from-behind view-point. However, the dynamics of the following vehicle sometimes change according to the driving condition. So, the robust platoon control system should be constructed to suppress the change of the condition. Fig.3 shows a block diagram of the driving control system. Here, obstacle avoidance in left side of the figure is discussed in a later section.

A. Robust control block using MEC

If the structure of the controlled object and its parameters are obtained accurately, the controller can be easily designed. As there is a parameter error or unmodeled dynamics, the error exists between the model and the actual plant. In the previous research, a model error compensator (MEC) specialized for suppression of the error between the plant and the model has been proposed [7]. The error occurs between the actual output and the ideal output of the reference model, as a result, control performance may degrade. Detail discussion is omitted but the I/O characteristics close to P_{Mv} and $P_{M\omega}$ by a robust controller. Table.I shows the parameters of the virtual vehicle and the reference model of MEC.

$$v_p = P_{Mv} J_y = \frac{1}{ms} J_y \tag{1}$$

$$\omega_p = P_{M\omega} J_x = \frac{K_\omega}{\tau s + 1} J_x \tag{2}$$

$$: \quad K_\omega = -\frac{d_f}{2ml_f v_p}, \ \tau = -\frac{I}{ml_f v_p}$$

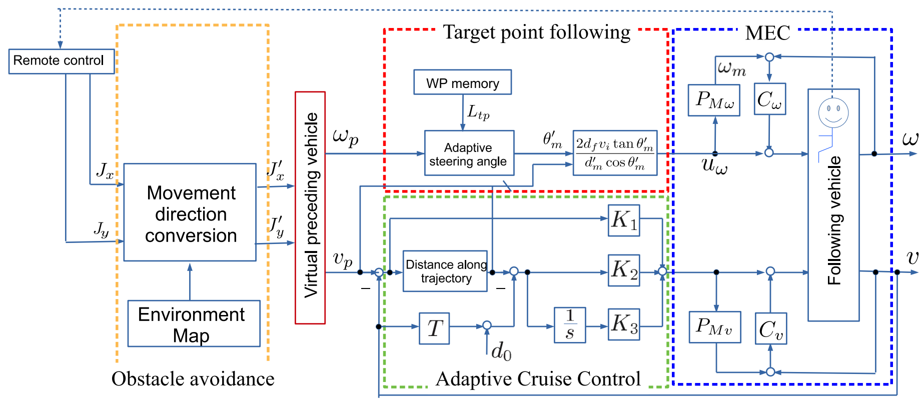


Fig. 3. Block diagram of virtual platoon control system

TABLE I
VIRTUAL VEHICLE PARAMETERS

m	vehicle body	145kg
I	inertia of body	54.25kgm^2
d_f	tread	0.62m
l_f	distance of COG	-0.2m
v_p	reference velocity	0.5m/s

B. Adaptive Cruise Control and Target Point Following blocks

The inter-vehicle distance d_m between the virtual and the following vehicles and the direction θ_m are expressed as follows,

$$d_m = \sqrt{(x_p - x_i)^2 + (y_p - y_i)^2} \tag{3}$$

$$\theta_m = tan^{-1}\frac{y_p - y_i}{x_p - x_i} - \theta_i \tag{4}$$

where $(x_i,\ y_i)$ and θ_i are the position and the angle of following vehicle.

The virtual preceding vehicle is remote-controlled from the following vehicle looking-from-behind view-point, and the vehicle followed the preceding one accurately. In this paper, Adaptive Cruise Control (ACC) algorithm and Target Point Following (TPF) algorithm with waypoints (WPs) are used for the virtual platoon control [4].

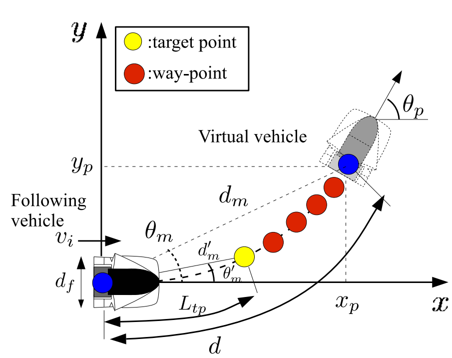


Fig. 4. Trajectory using TPF algorithm on 2D plane

ACC controls the inter-vehicle distance by controlling the relative speed of vehicles. The detail discussion is omitted, but the target inter-vehicle distance d_r is defined as following equation according to the driving speed v_i.

$$d_r = Tv_i + d_0 \tag{5}$$

where $T = 2.0$s is the target inter-vehicle time. v_i is the following vehicle speed and $d_0 = 2.0$m is the stopping inter-vehicle distance.

Target Point Following (TPF) algorithm controls the steering angle so that the vehicle travels along a trajectory of the preceding vehicle [8]. Using TPF algorithm, the desired steering angle is calculated so as that the vehicle's position passes with the desired attitude angle. It is assumed that the position and the angle are $(x_p,\ y_p)$ and θ_p. WPs are history of the positions of the virtual vehicle and the tracking errors can be reduced by using targeting variable waypoint at a curve [4].

In Fig.4, the inter-vehicular distance d is calculated along wheel track marked by WPs. The attitude angle θ'_m and the inter-vehicle distance d'_m between the WPs and the following vehicles are changed adaptively according to the curvature. The steering command u_ω is approximated as follows,

$$u_\omega \approx \frac{2d_f v_p \tan\theta'_m}{d'_m \cos\theta'_m}. \tag{6}$$

C. Remote Control block with 3D motion of a virtual vehicle

As the position/attitude of the vehicle is defined on the floor, the real time translation to the view point of the driver is required . In order to convert the 3D image on the floor into driver's 2D image on HMD, a model-view, a projection and a viewpoint transformations are required. A flowchart of the conversion process is summarized in Fig.5 [5].

Fig.6 shows a virtual vehicle on HMD created by these processes. As a virtual vehicle is displayed by the footprint, that is the bottom surface of the vehicle, the blind area of the driver does not occur at the projection of preceding vehicle.

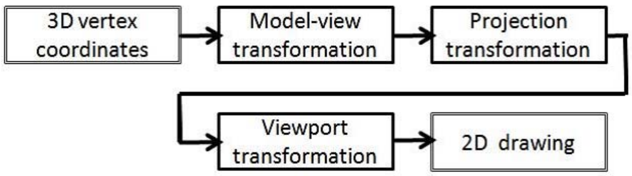


Fig. 5. Block diagram of 3D translation

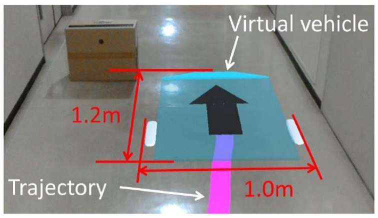


Fig. 6. Image of virtual vehicle

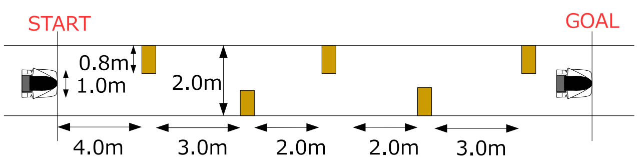
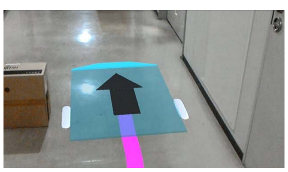
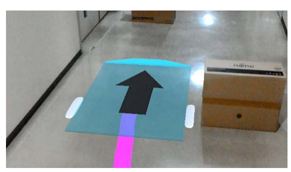


Fig. 7. Arrangement of obstacles in a narrow corridor

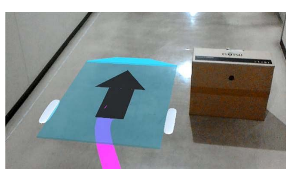


(a)1st obstacle (4m)



(b)2nd obstacle (7m)

(c)3rd obstacle (10m)



(d)4th obstacle (13m)

Fig. 8. Experimental view of virtual vehicle at obstacles

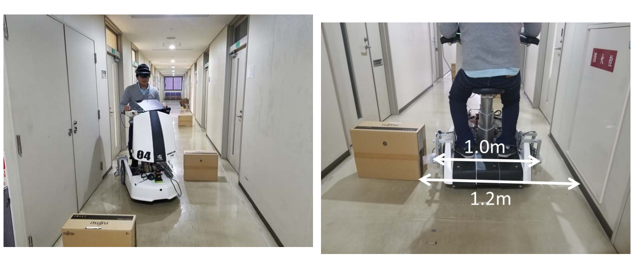


Fig. 9. Experimental photograph

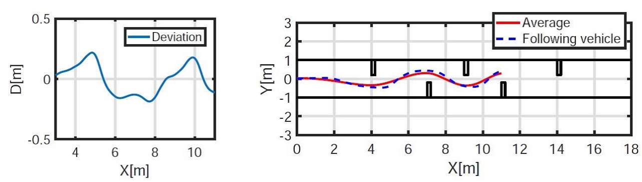
IV. Preliminary Experiment using Assist System by Beginner

In order to evaluate the fundamental performance of the driving assistance, the vehicle drives a narrow corridor where obstacles are placed. The layout of obstacles was shown in Fig.7. As the clearance between the vehicle and the obstacle is 0.2m, it is difficult to steer the vehicle well for beginners. It is remarked that the interval is narrowing from 2nd to 4th obstacles. As a result, the difficulty of driving increased.

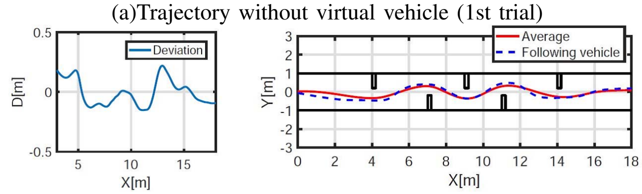
Fig.8 show the images nearby obstacles on the HMD. The blue thin board is the virtual vehicle and the magenta line means the trajectory that the following vehicle should be passed. The maximum speed of the virtual vehicle is 0.5m/s (about 2km/h), the maximum yaw rate is 0.5rad/s, $T = 2.0$s, $d_0 = 2.0$m.

Fig.10 shows trajectories of a beginner and its deviation from expert. Here, a beginner is a driver within 10 minutes of boarding time, and an expert is a driver with a boarding time of 20 hours or more. Red line means the average of the trajectories by the expert, and dotted line means the trajectory of the welfare vehicle.

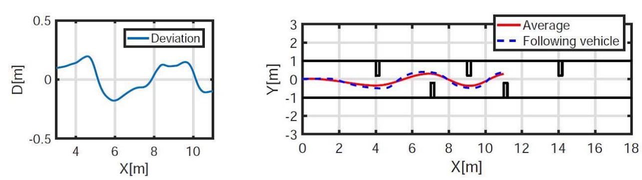
Fig.10(a) was a first trial without driving assist (normal drive), and the vehicle collided with the wall at 4th obstacle. At 2nd trial, the vehicle collided with the wall just the same. And



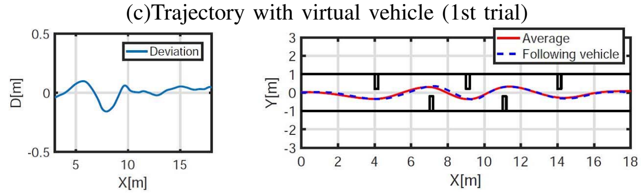
(a)Trajectory without virtual vehicle (1st trial)



(b)Trajectory without virtual vehicle (3rd trial)



(c)Trajectory with virtual vehicle (1st trial)



(d)Trajectory with virtual vehicle (3rd trial)

Fig. 10. Drive trajectory by a beginner

the vehicle reached the goal at 3rd trial as shown in Fig.10(b). On the other hand, the vehicle with the driving assistant system reached the goal at 2nd trial. Fig.10(d) shows the trajectory at 3rd trial.

Next, the deviations between expert's and beginner's trajectories were focused in order to evaluate the quality of driving skill. Fig.11 shows the sum of squared errors where reference average trajectory is defined by average of 5 times travel by an expert. The square error of deviation is evaluated at 3m $\sim$ 11m focusing on the driving in narrow space. In normal driving without assist, the square error of the deviation decreased slightly, but its value was relatively large. Also, the squared error using proposed method was very small, when the vehicle reached the goal.

It is remarked that the most of failure by the beginner is collision with the wall. So, if the support of avoidance from the wall can be available, an advanced safety will be expected.

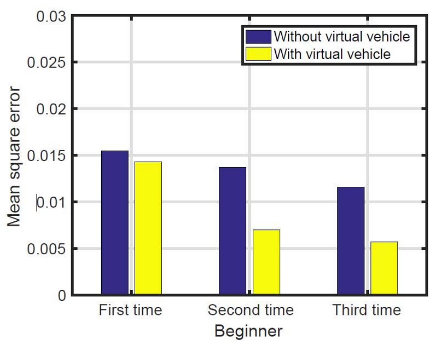


Fig. 11. Mean squared error from expert trajectories

V. Experimental Drive to avoid collision to the wall using Mixed Reality

The HoloLens used in the experiment has the 3D mapping function of running environment as shown in Fig.12. For example, Fig.13(a) is a 3D model of all corridor. The green zone in Fig.13(b) shows the safety driving area on the floor. Here, the safety driving zone can be obtained assuming that the virtual vehicle is a cylinder with a radius of 0.6m using search algorithm.

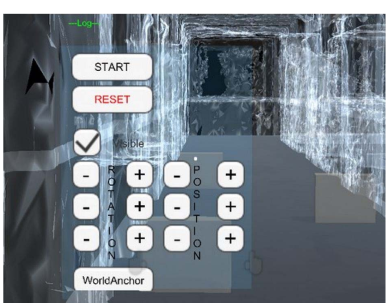
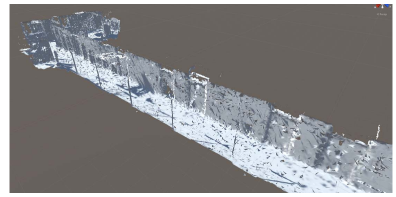
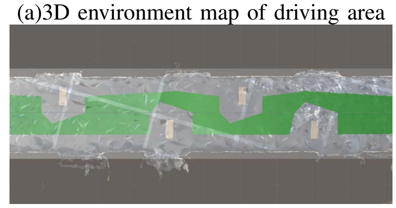


Fig. 12. 3D mapping of corridor by HoloLens



(a)3D environment map of driving area



(b)Safety area on floor

Fig. 13. 3D environment map and safety driving area

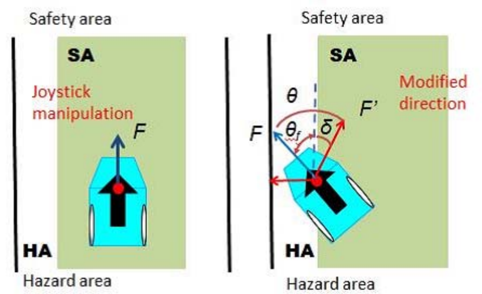


Fig. 14. Modification of joystick operation for virtual vehicle

A wall collision avoidance function of proposed block diagram is shown in Fig.3. When the virtual vehicle seeks to drive into the hazardous area (HA), the joystick operation is modified to be corrected so as to change the movement direction into safety area (SA). This algorithm to modify the joystick manipulation is describes as follows,

$$F' = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} F \tag{7}$$

$$: \theta = \begin{cases} 0 & \in SA \\ \phi & \in HA \end{cases} \tag{8}$$

where F is joystick control vector $F = [J_x,\ J_y]^T$, and F' is its modified vector. The modified angle is $\phi = \theta_f + \delta$ in HA, where θ_f is an angle between F and the corridor, δ is an offset angle to return the virtual vehicle to SA. As shown in Fig.14, the control vector F of the virtual vehicle is modified F' so as to return to the safety area. Using this algorithm, if the virtual vehicle enters HA, the misdirected joystick command is modified and get back to SA. After then, the virtual vehicle is kept in SA without modification by the driver.

In Fig.15 , as the expert drove the virtual vehicle well in the safe region. The operation was not modified by the proposed method. A trajectory of following vehicle (STAVi) was shown in Fig.16. In case of the beginner, the virtual vehicle entered into the hazardous area in 3th and 4th obstacles. The operation

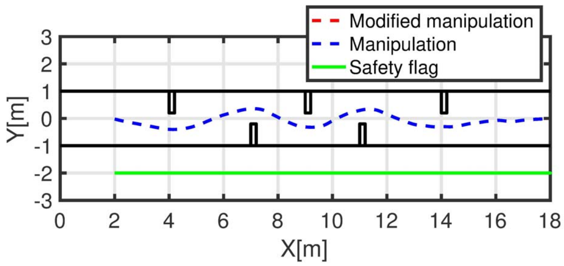


Fig. 15. Trajectory of virtual vehicle using proposed method (expert)

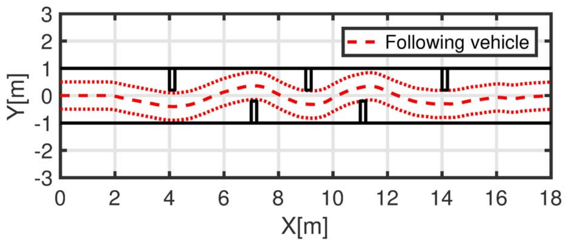


Fig. 16. Trajectory of following vehicle using proposed method (expert)

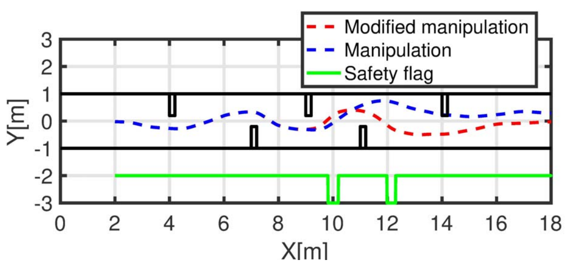


Fig. 17. Trajectory of virtual vehicle using proposed method (beginner)

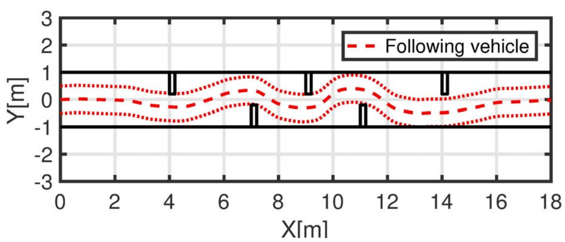


Fig. 18. Trajectory of following vehicle using proposed method (beginner)

was modified in 10s and 12s as shown in Fig.17 where the green line in the figure means a collision flag. As a result, a trajectory of the following vehicle was modified well as shown in Fig.18. Fig.19 shows modified steering rate ω of STAVi. From the figure, it was observed that the steering command was modified in in 10s and 12s.

VI. Conclusion

In this paper, the driving assistance using a virtual platoon control from an simulated objective viewpoint was evaluated

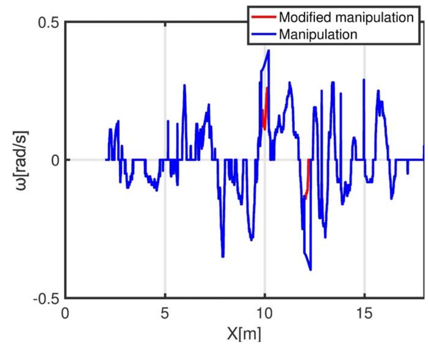


Fig. 19. Modified joystick operation (beginner)

by experiments for beginners. In experiments, although the trajectory collides with obstacles, it gets closer to the expert's trajectory every time. It was confirmed that proposed assistance system is effective for the beginner, however, a most of driving failure of a beginner was a collision with the wall.

So, using MR technology, a method to avoid static obstacle avoidance using 3D modeling of the static environment was proposed. The assistance system with collision avoidance with the wall was designed using the 3D-data of the driving environment by HMD and the collision avoidance ability was evaluated. As a result, a beginner drove the STAVi without getting a collision.

The avoidance system proposed in this paper marks the beginning of MR application for the welfare. More safety system with HMD should be designed in many driving contexts in the future. Since it is difficult to evaluate driving skills and accustoming, the evaluation using statistical processing with many drives is also the future work. This work was supported by JSPS KAKENHI Grant Number JP17K06231 and Tateisi Science and Technology Foundation.

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